Random Forest — Feature Set

Input features of the saved multi-output Random Forest (labels: CDI stage + SPI-3 stage - CDI/VCI are never features)

Feature	Group	Source	Description
SPI-1	Meteorological (Open-Meteo)	Open-Meteo archive, single point	1-month standardized precipitation index (Gamma fit, 1991-2020)
SPI-3	Meteorological (Open-Meteo)	Open-Meteo archive, single point	3-month standardized precipitation index - primary drought signal
SPI-6	Meteorological (Open-Meteo)	Open-Meteo archive, single point	6-month standardized precipitation index (hydrological memory)
SPI-12	Meteorological (Open-Meteo)	Open-Meteo archive, single point	12-month standardized precipitation index (long-term drought)
SSMI	Soil moisture (SMAP)	NASA SMAP L4 SPL4SMGP (GEE), ~11 km	Soil-moisture index standardised per tehsil (z-score, mean ~0)
NDVI	Optical vegetation (Sentinel-2)	Sentinel-2 GEE composites, 100 m	Normalized difference vegetation index, monthly cloud-masked composite
EVI	Optical vegetation (Sentinel-2)	Sentinel-2 GEE composites, 100 m	Enhanced vegetation index, monthly cloud-masked composite
PCI	Optical vegetation (Sentinel-2)	Sentinel-2 GEE composites, 100 m	Precipitation condition index derived from NDVI/EVI climatology

Leakage caveat: the CDI-stage label is itself built from SPI-3/SSMI/NDVI(EVI)/EVI/PCI, which are also RF features - high accuracy is therefore expected (see fig_model_validation).